



Lab 07: VLAN Segmentation

2960-24TT Switch · PC0 (VLAN 10) · PC1 (VLAN 20) · PC2 (VLAN 30)

Lab #07

Topic: VLANs & Switching

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Create and name three VLANs (SALES, TECH, MGMT) on a single 2960-24TT switch.
- Assign PC0 to VLAN 10, PC1 to VLAN 20 and PC2 to VLAN 30 via access ports.
- Confirm that PC0 and another device on VLAN 10 can communicate at Layer 2.
- Confirm that cross-VLAN pings fail — demonstrating correct VLAN isolation.
- Relate to Network+ objective 2.3: Purpose and use of VLANs.

2. Network Topology

A single 2960-24TT switch (Switch0) with no router attached. PC0 on Fa0/1 (VLAN 10 – SALES), PC1 on Fa0/2 (VLAN 20 – TECH), PC2 on Fa0/3 (VLAN 30 – MGMT). Without a Layer 3 device, all cross-VLAN pings will fail — this is the expected and correct behaviour.

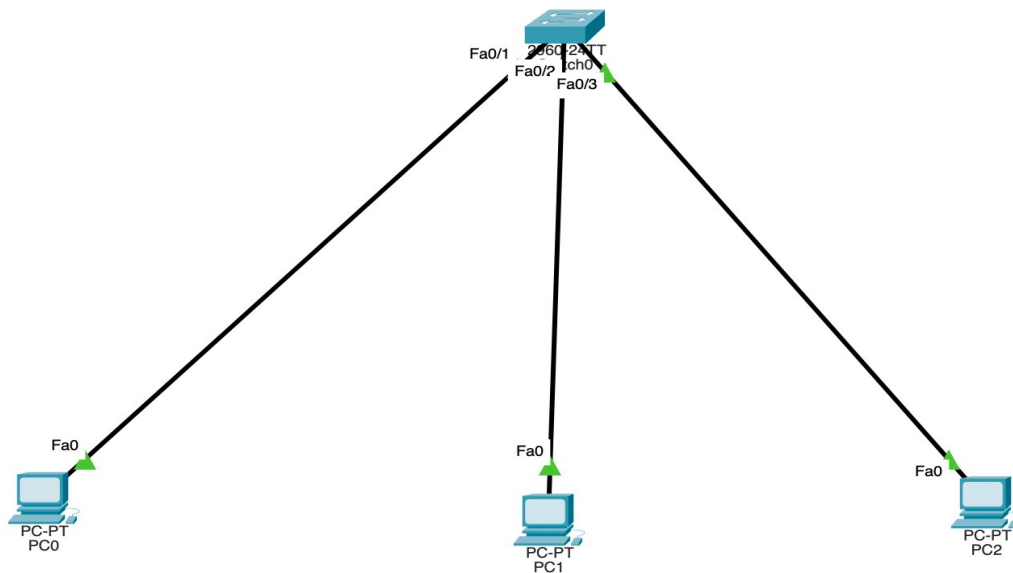


Figure 1 – Single Switch0 with PC0, PC1, PC2 on three separate VLANs

3. IP Addressing Table

Device	Switch Port	VLAN ID	VLAN Name	IP Address	Subnet Mask
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PC0	Fa0/1	10	SALES	192.168.10.10	255.255.255.0
PC1	Fa0/2	20	TECH	192.168.20.10	255.255.255.0
PC2	Fa0/3	30	MGMT	192.168.30.10	255.255.255.0

4. Step-by-Step Configuration

4.1 Create VLANs on Switch0

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name TECH
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name MGMT
Switch(config-vlan)#exit
```

4.2 Assign Access Ports

```
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch#write memory
```

5. Verification & Expected Results

Run the commands below and check whether each ping succeeds or fails as expected:

Command	Expected Result	Reason
Switch#show vlan brief	Lists all 3 VLANs with ports	Confirms correct VLAN assignment



PC0>ping 192.168.10.10	Reply (success)	Same VLAN 10 — Layer 2 forwarding
PC0>ping 192.168.20.10	Request timeout (fail)	Different VLAN — no routing
PC0>ping 192.168.30.10	Request timeout (fail)	Different VLAN — no routing

6. Key Takeaways

- VLANs divide a single physical switch into multiple logical broadcast domains.
- Access ports carry traffic for ONE VLAN only — frames leave and arrive untagged.
- Cross-VLAN isolation without a router is the correct expected behaviour — not an error.
- VLANs improve security, reduce broadcast traffic and simplify network management.
- **show vlan brief** is the fastest command to verify port-to-VLAN assignments.